Chapter 11 — Special Theory of Relativity

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. Special relativity applies to:

(a) only rotating frames **(b)** only gravitational frames **(c)** only accelerating frames **(d)** inertial frames

2. An inertial frame is one where:

(a) objects always accelerate **(b)** time is absolute **(c)** light speed changes **(d)** Newton's first law holds

3. The speed of light in vacuum is:

(a) less for moving observer **(b)** zero **(c)** same for all observers **(d)** greater for moving source

4. The laws of physics are:

(a) only valid at rest **(b)** only valid on Earth **(c)** different in moving frames **(d)** same in all inertial frames

5. Time dilation means moving clocks run:

(a) backwards **(b)** slow **(c)** fast **(d)** same

6. Proper time is measured by:

(a) stationary observer only **(b)** any clock **(c)** moving clock **(d)** clock at rest relative to events

7. Length contraction occurs:

(a) along motion **(b)** in all directions **(c)** perpendicular to motion **(d)** only at rest

8. Proper length is measured when object is:

(a) accelerating **(b)** rotating **(c)** moving **(d)** at rest relative to observer

9. Relativistic mass is:

(a) m_0v/c **(b)** $m_0/\sqrt{(1 - v^2/c^2)}$ **(c)** $m_0\sqrt{(1 - v^2/c^2)}$ **(d)** m_0c^2

10. As v approaches c, relativistic mass:

(a) increases without limit **(b)** decreases **(c)** becomes zero **(d)** remains m_0

11. A material body cannot reach c because:

(a) it has no mass **(b)** infinite energy would be needed **(c)** light stops it **(d)** time stops

12. Mass–energy relation is:

(a) $E = m/c^2$ **(b)** $E = mc^2$ **(c)** $E = mc$ **(d)** $E = \frac{1}{2}mc^2$

13. Rest energy is:

(a) $\frac{1}{2}m_0v^2$ **(b)** m_0gh **(c)** mc **(d)** m_0c^2

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14. Relativistic kinetic energy is:

(a) $(m - m_0)c^2$ **(b)** $\frac{1}{2}mv^2$ always **(c)** mc^2 **(d)** m_0c^2

15. If rest mass increases to m, kinetic energy is:

(a) $(m - m_0)c^2$ **(b)** mc^2 **(c)** $\frac{1}{2}mc^2$ **(d)** $\frac{1}{2}(m - m_0)c^2$

16. The speed of light from a moving car is:

(a) less **(b)** greater **(c)** zero **(d)** same as at rest

17. A photon has rest mass:

(a) zero **(b)** 9.1×10^{-31} kg **(c)** 1.67×10^{-27} kg **(d)** infinite

18. Relativistic effects are noticeable at:

(a) zero speed **(b)** sound speed **(c)** everyday speeds **(d)** speeds close to c

19. Space-time has:

(a) two dimensions **(b)** four dimensions **(c)** three dimensions **(d)** one dimension

20. In general relativity, gravity is:

(a) an electric force **(b)** curvature of space-time **(c)** a contact force **(d)** a magnetic force

21. Bodies in curved space-time follow:

(a) straight lines in space only **(b)** circles always **(c)** parabolas always **(d)** geodesics

22. The 1919 eclipse confirmed:

(a) photoelectric effect **(b)** bending of starlight **(c)** mass increase only **(d)** time dilation only

23. Simultaneity is:

(a) absolute **(b)** always same **(c)** relative **(d)** not measurable

24. If v = 0.8c, the factor $\sqrt{(1 - v^2/c^2)}$ is:

(a) 1 **(b)** 0.8 **(c)** 0.4 **(d)** 0.6

25. If v = 0.6c, the factor $\sqrt{(1 - v^2/c^2)}$ is:

(a) 1 **(b)** 0.6 **(c)** 0.8 **(d)** 0.4

26. If v = 0.99c, length contracts to about:

(a) 0.14 L_0 **(b)** 0.5 L_0 **(c)** 0.99 L_0 **(d)** L_0

27. If v = 0.8c, dilated time is:

(a) 0.6 t_0 **(b)** 1.67 t_0 **(c)** 2 t_0 **(d)** t_0

28. If mass triples, speed is about:

(a) 0.8c **(b)** 0.943c **(c)** 0.6c **(d)** 0.99c

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29. Classical mechanics is a good approximation when:

(a) $v > c$ **(b)** $v \approx c$ **(c)** $v \ll c$ **(d)** $v = c$

30. Relativity does not contradict Newton's laws because:

(a) they are wrong always **(b)** they are low-speed limits **(c)** they have no mass **(d)** they apply only to light

31. If c were infinite, relativistic equations reduce to:

(a) infinite mass **(b)** quantum equations **(c)** zero energy **(d)** classical equations

32. The ultimate speed limit for material objects is:

(a) escape velocity **(b)** c **(c)** orbital speed **(d)** sound speed

33. Proper time is always:

(a) zero **(b)** shorter than dilated time **(c)** same as dilated time **(d)** longer than dilated time

34. A moving rod appears:

(a) wider **(b)** longer along motion **(c)** heavier only **(d)** shorter along motion

35. The rest energy of electron is about:

(a) 1.02 MeV **(b)** 0.51 MeV **(c)** 931 MeV **(d)** 125 GeV

36. Pair production minimum photon energy is:

(a) 125 GeV **(b)** 931 MeV **(c)** 0.51 MeV **(d)** 1.02 MeV

37. Electron–positron annihilation produces photons of:

(a) 0.51 MeV each **(b)** 931 MeV each **(c)** 125 GeV each **(d)** 1.02 MeV each

38. Relativistic momentum is:

(a) mc **(b)** $\frac{1}{2}mv^2$ **(c)** m_0v always **(d)** mv with relativistic mass

39. The space-time interval is:

(a) invariant **(b)** always time only **(c)** always zero **(d)** always space only

40. Gravitational waves are ripples in:

(a) magnetic field only **(b)** space-time **(c)** air **(d)** water

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Answer Key with Brief Reasons

1. Answer: (d) inertial frames — It deals with uniform motion.

2. Answer: (d) Newton's first law holds — No net force means constant velocity.

3. Answer: (c) same for all observers — Second postulate.

4. Answer: (d) same in all inertial frames — First postulate.

5. Answer: (b) slow — $t = t_0/\sqrt{(1 - v^2/c^2)}$.

6. Answer: (d) clock at rest relative to events — It is the shortest time.

7. Answer: (a) along motion — Only parallel dimension contracts.

8. Answer: (d) at rest relative to observer — Rest length is longest.

9. Answer: (b) $m_0/\sqrt{(1 - v^2/c^2)}$ — Mass increases with speed.

10. Answer: (a) increases without limit — Denominator approaches zero.

11. Answer: (b) infinite energy would be needed — Mass grows without limit.

12. Answer: (b) $E = mc^2$ — Einstein relation.

13. Answer: (d) m_0c^2 — Energy of rest mass.

14. Answer: (a) $(m - m_0)c^2$ — Total minus rest energy.

15. Answer: (a) $(m - m_0)c^2$ — $K = E - E_0$.

16. Answer: (d) same as at rest — Light speed is invariant.

17. Answer: (a) zero — Photon rest mass is zero.

18. Answer: (d) speeds close to c — v^2/c^2 must be significant.

19. Answer: (b) four dimensions — Three space + one time.

20. Answer: (b) curvature of space-time — Mass curves space-time.

21. Answer: (d) geodesics — Straightest paths.

22. Answer: (b) bending of starlight — General relativity test.

23. Answer: (c) relative — Different frames disagree.

24. Answer: (d) 0.6 — $\sqrt{(1 - 0.64)} = 0.6$.

25. Answer: (c) 0.8 — $\sqrt{(1 - 0.36)} = 0.8$.

26. Answer: (a) 0.14 L_0 — $\sqrt{(1 - 0.9801)} \approx 0.14$.

27. Answer: (b) 1.67 t_0 — $1/0.6 \approx 1.67$.

28. Answer: (b) 0.943c — $\sqrt{(1 - 1/9)} \approx 0.943$.

29. Answer: (c) $v \ll c$ — Relativistic factor ≈ 1.

30. Answer: (b) they are low-speed limits — Newtonian results emerge at small v.

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31. Answer: (d) classical equations — No relativistic corrections.

32. Answer: (b) c — Mass increases without limit.

33. Answer: (b) shorter than dilated time — Moving clocks run slow.

34. Answer: (d) shorter along motion — Length contraction.

35. Answer: (b) 0.51 MeV — $m_e c^2 \approx 0.511$ MeV.

36. Answer: (d) 1.02 MeV — Two electron rest energies.

37. Answer: (a) 0.51 MeV each — Rest energy shared by two photons.

38. Answer: (d) mv with relativistic mass — $p = mv$.

39. Answer: (a) invariant — All observers agree on it.

40. Answer: (b) space-time — They distort space-time.

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